



from the labs

ESP

The labs at MIT are also working on Emotional Social Intelligence Prosthetic, a device for reading minds.

outwards on the phone's screen with two fingers. In SixthSense, the same is done by moving all four fingers, or the hands, outwards. And vice versa: bring the hands in to zoom out.

...an Indian angle

One of the coolest aspects of the iPhone was its 'one-button-to-rule-them-all', that let you go to the home screen from within any application. It made navigation so much simpler in an interface devoid of a physical keyboard and a mouse, with their many shortcuts.

So how does Pranav 'welcome' a user to his home screen? The Indian connection comes to the fore, with the folded-hands "Namaste" gesture serving the purpose — from within any application, just like the iPhone.

It seems so simple, it's almost as if the MIT team was inspired by a game of dumb charades. In fact, freehand gestures, or postures, are just that — signs used in everyday life to depict certain events or actions. The demonstration video aptly describes many of these easy-to-understand gestures.

Pranav walks around a city, clicking pictures of various sites. He doesn't have a digital camera with him, but why would he need one? He simply lifts up his two hands, and joins the forefinger tips of one hand to the thumb tips of another — forming a frame. Once he "frames" a scene, the camera takes over and clicks a picture, which is saved for viewing later.

Myriad applications

Indeed, the uses of this technology are limited only by one's imagination.

Pranav doesn't wear a wristwatch any more. He simply looks at the back of his hand, and draws a circle near the wrist. Like with the number pad, a watch pops up out of nowhere!

"When you go to the supermarket, and you're standing there in that huge aisle of different types of toilet paper, you don't take out your cellphone and open a browser and go to a web site to try to decide which of these is the most ecologically responsible decision to make," Maes said during her address at TED.

In her demonstration video, Pranav walked down such an aisle, and picked up a pack of paper towels. The SixthSense projected an unfavourable ecological rating on it, and Pranav moved to the next pack. This one got a green light, and the engineer could even tap that to gain some more information about the product. What a satisfactory and knowledgeable way to make a purchase!

And Maes stressed how customisable SixthSense is: "Some of you may want the toilet paper with the most bleach in it rather than the most ecological choice."

Another scenario had the 'designeer' looking at his airplane ticket. His pendant locked on to the little piece of paper, and alerted him about any delays in schedule and changes in the designated gate.

The most interesting of the aspects — especially for newspaper and magazine readers — was the use of the augmented reality tool with an article. On a blank piece of paper next to the article, or even a white space, the SixthSense can project relevant videos, news channel feeds, updated scores of matches, etc.

A controversial use of the device had Pranav meeting his colleague, Chang. On Chang's white t-shirt, the SixthSense projected "world clouds" linked to him across the internet, creating a virtual tag-based Google search result of his online persona. Everything from his hobbies to work was there to be seen — an aspect that could throw up many privacy issues in the future.

A fifth sense for the disabled

More than anything, an issue close to Pranav's heart is the application of his technology back home.

"I think the SixthSense and the vision behind it can act as the gap-filler for the digital divide in India and other developing nations," he says. "I am really excited to extend the SixthSense

vision to the fields of education, medicine and accessibility."

The engineer is currently working on versions of the SixthSense that can act as an eye for the blind and an ear for the deaf; or it can speak on behalf of a person who cannot speak. "It could act as the missing fifth sense, rather than a sixth sense," he enthuses.

Pranav also hopes to get kids back into playgrounds with his device: "I want to dig into education for kids, such that the power of digital world can be integrated into their everyday activities, including playing. Today's kids are missing out on playing in real fields. I think the underlying technologies in SixthSense can redefine the world of education and gaming drastically, by making the world a playground."

What Now And What Next?

The potential for the technology was evident at TED, an event that hosts some of the biggest minds in the world today — no exaggeration. At the end of Maes' speech, Pranav stood up to receive a standing ovation from a crowd that has seen and regularly participates in creating technologies that promise to change our world.

The commercially-available and off-the-shelf components hold a lucrative prospect in the near future: "In mass production, the device shouldn't cost more than today's cellphones," Maes said.

The elimination of physical input hardware makes it easy for the device to be scaled down in size in the future — opening the door for a more stylish product.

"I think it will take about 1.5-2 years before this can go into market," says Pranav, who is working closely with several consumer electronics companies. Already, Pranav and the team are planning several improvements. The topmost among these is the development of a more sophisticated gesture-recognition technique that rids the user of the cumbersome need of wearing coloured markers.

"In a couple of months, we will be ready with a version that will eliminate the requirement of the coloured markers," he says.

"I am also making some changes in the system design so that one can remove the SixthSense pendant, put it on any table, and start using that table as an interactive multi-touch screen," Pranav says.

"It's not perfect yet, and it's still a work in progress," Maes adds. And then the mad scientist took over: "Who knows, in ten years, we might come up with the ultimate 'sixth sense' brain implant." ■

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Here is a brief on the computer vision part of SixthSense. According to Mistry, "SixthSense continuously captures and processes the video stream captured by the camera. The core software component of SixthSense is based on amalgamation of several machine vision algorithms."

"SixthSense system is capable of tracking and recognizing the user's hand movements and freehand gestures. A technique that I have developed using edge detection and color tracking helps performing this operation. The system also recognises and tracks the location of physical objects in users viewpoint. The system uses a few feature extraction and image matching algorithms in order to compare and identify pictures of the objects with the objects in user's viewpoint. The system later retrieves related information about the identified object from the Internet and projects on the object itself, seamlessly augmenting them. SixthSense also is able to stabilise and correct the projected visual information on non-flat surfaces and objects. At the moment, I am also working on some additional features that will correct(change) the interface colors real-time on the basis of the characteristics of the surface the information need to be projected. All these algorithms are developed using C++ and C# and are based mainly on machine-vision based techniques. The visual interface is programmed using WPF," he added.